

Correction of OCR results using large language models

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ABSTRACT

This paper proposes a large language model (LLM)-based method for correcting OCR results, integrating prompt engineering with recognition models including PP-OCRv4 and GLM-4-Flash. By introducing an LLM-driven error correction mechanism, the accuracy of OCR-recognized texts is significantly improved, particularly in punctuation restoration and character rectification. Experimental results demonstrate that for datasets with GLM correction confidence scores below 0.95, the proposed method achieves an accuracy improvement of approximately **17%**. Specifically, the GLM-enhanced framework outperforms standalone PP-OCRv4 by **4.7%** and surpasses the Kimi large model by **29.5%** in accuracy. These findings validate the high practical applicability of the GLM-based correction strategy in real-world scenarios.

Keywords: OCR; large language model; PaddlePaddle; GLM; Kimi; prompt engineering

1. INTRODUCTION

Optical Character Recognition (OCR) has transformed data extraction from printed and handwritten texts, enhancing efficiency in document management, healthcare, and archival services. Nonetheless, OCR still faces challenges with low-quality images, complex layouts, multilingual texts, and unconventional fonts, leading to errors such as misrecognized characters and misplaced punctuation that compromise data accuracy in critical applications like legal and medical records.

Traditional error correction methods based on rule-based or statistical models have limited ability to handle complex linguistic contexts. Recent advances in deep learning and large language models (LLMs) (e.g., BERT, GPT, and GLM) offer improved correction by leveraging extensive datasets and fine-tuning on domain-specific data, thereby better addressing spelling, punctuation, and syntax errors.

In this paper, we compare conventional techniques with modern LLM-enhanced frameworks and propose a hybrid correction architecture. Our approach integrates PP-OCRv4 for initial text recognition with a post-processing pipeline that combines prompt engineering and GLM-based refinement. Experimental results demonstrate that our method significantly improves OCR accuracy and robustness while maintaining scalability for large and heterogeneous datasets.

2. RELATED WORK

Recent advancements in OCR technology have expanded its applications across industries, driving increased attention to post-OCR correction methodologies. Current approaches can be categorized into four main paradigms: rule-based, statistical, traditional machine learning (ML)-based, and deep learning-based methods.

2.1 Rule-Based methods

Rule-based approaches employ manually designed templates or linguistic rules (e.g., spelling checkers, domain-specific text patterns) to rectify OCR errors. While widely adopted in early-stage research, these methods require substantial human effort for rule creation and exhibit limited generalization capabilities in complex textual scenarios^[1].

2.2 Statistical methods

Statistical techniques leverage language models (e.g., n -gram analysis) to correct OCR outputs by estimating word/sentence likelihoods based on textual corpora^[2,3]. These methods demonstrate effectiveness in standardized text correction, particularly for spelling and high-frequency word errors. However, they struggle with semantic ambiguities

and contextual dependencies. Performance further degrades when training data diverges from target text domains.

2.3 Traditional machine learning methods

ML-based frameworks formulate OCR correction as classification tasks using engineered features (e.g., character-level errors, punctuation patterns). Algorithms such as Support Vector Machines (SVM) and Random Forests [4-6] improve adaptability across text types compared to rule-based systems. Nevertheless, their reliance on manual feature engineering limits scalability for handling syntactic complexities.

2.4 Deep learning methods

The integration of deep learning into NLP has revolutionized OCR correction, primarily through two architectures:

2.4.1 Sequence-to-sequence models

Seq2seq models (e.g., LSTM, Transformer) map OCR-recognized text to corrected sequences [7,8]. These models excel at capturing contextual and grammatical relationships, enabling automated long-text correction without feature engineering.

2.4.2 Large language models (LLMs) with prompt engineering

Pretrained LLMs (BERT, GPT, GLM) combined with prompt-based strategies have demonstrated state-of-the-art performance in semantic error correction and cross-domain adaptation. By designing task-specific prompts (e.g., "Correct OCR errors in the following text: [INPUT]"), these frameworks effectively guide LLMs to resolve contextual ambiguities, showing particular promise for multilingual and structurally complex documents.

3. METHODOLOGY

This section details our framework, dataset characteristics, and implementation specifics of PP-OCRv4, GLM-4-Flash, GLM-4-Plus, and Kimi models.

3.1 Chinese novel dataset

The dataset derives from Jia Li's Diary by contemporary Chinese children's literature author Qin Wenjun. We generated 1 million synthetic images (totaling 3.26 million characters) by applying randomized variations in font type, size, grayscale, blur, perspective distortion, and stretching. Each image contains randomly cropped sentences with mixed Chinese/English punctuation and variable character counts.

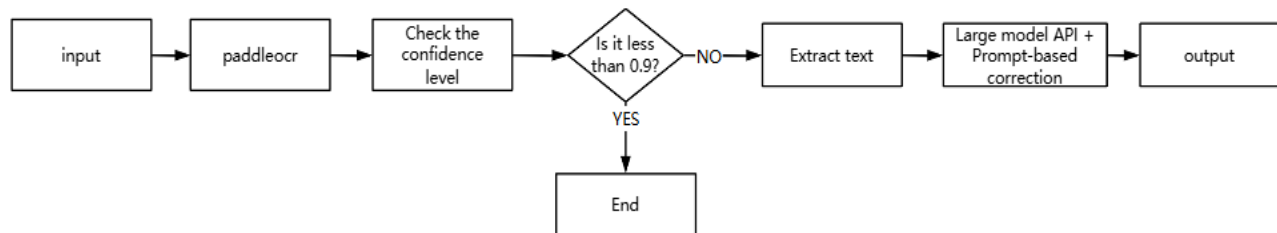


Figure 1. Experimental Process.

Figure 1 illustrates the workflow. We employed PaddlePaddle's lightweight PP-OCRv4 model to process all images, generating text outputs with confidence scores. Samples scoring below 0.95 confidence threshold (276,167 instances) were flagged for subsequent LLM-based correction.

3.2 PP-OCRv4 architecture

PP-OCRv4 is the latest OCR system from the PaddleOCR team, built on PP-OCRv3 with key optimizations. While the overall architecture remains the same, enhancements have been made to the detection and recognition models, including improvements in data, network structure, and training strategies. The system framework is shown below (**Figure 2**).

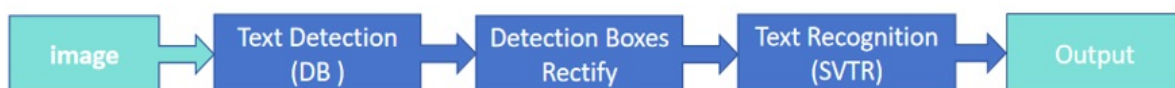


Figure 2. The system framework of PP-OCRv4

3.3 GLM and Kimi LLM Integration

GLM-4-Flash (Zhipu AI): A high-throughput LLM supporting 1,500 concurrent requests, achieving 72.14 tokens/sec generation speed. Key features include: Real-time web retrieval, Function calling APIs, Domain-specific fine-tuning.

GLM-4-Plus: A precision-tuned variant of GLM-4-Flash for complex semantic tasks.

Kimi (Moonshot AI): A GPT-style Transformer model specializing in NLP tasks (text generation, semantic parsing).

For correction, low-confidence PP-OCRv4 outputs were routed to GLM-4-Flash/Plus via API calls with structured prompts. Separately, 100 sample images were directly processed by Kimi for end-to-end OCR and prompt-based refinement (see **Figure 3**).

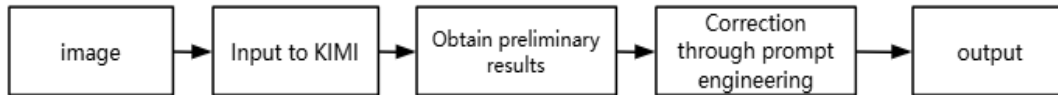


Figure 3. KIMI's second revision of data

3.4 Prompt engineering framework

We formalize OCR correction as a constrained language model generation task using the following template:

prompt = (

"I obtained this text via OCR, but it may contain punctuation or character errors. "

"As a professional text proofreader, please correct it while preserving the original text length.\n"

f"OCR text: {ocr_result.strip()}\n"

"Output only the corrected text without additional explanations.")

Design Rationale:

1. Role Specification ("professional text proofreader"): Explicitly guides the LLM to adopt an editorial persona, prioritizing grammatical fidelity.

2. Length Preservation ("preserving the original text length"):

Mitigates arbitrary insertions/deletions common in free-form LLM outputs.

3. Output Control ("only the corrected text"):

Suppresses explanatory text that could contaminate downstream pipelines.

3.4.1 Integration with OCR Pipeline

The hybrid correction workflow operates as shown in **Figure 4**.

```
ocr_output = PP_OCRv4(image)
if ocr_output.confidence < 0.95:
    corrected_text = GLM4_Plus.generate(prompt, temperature=0.2)
else:
    corrected_text = ocr_output.text |
```

Figure 4. Threshold-based Routing Mechanism between OCR and LLM Modules

4. RESULTS

After processing low-confidence OCR outputs with GLM-4-Flash, GLM-4-Plus, and Kimi LLMs, we systematically compared their corrected texts against ground-truth data. Key quantitative findings are summarized below.

4.1 Performance of GLM-4-Plus vs. GLM-4-Flash

The computational efficiency and correction accuracy of GLM-4-Plus and GLM-4-Flash are compared in **Table 1**.

Table 1. compares the computational efficiency and correction accuracy of GLM-4-Plus and GLM-4-Flash:

Metric	GLM-4-Plus	GLM-4-Flash
Model Parameters	130B	7B
Accuracy Improvement	+4.7% (vs. baseline)	+1.56% (vs. baseline)
Tokens/sec	28.5	72.14
Latency per Request	1.8s	0.7s
Punctuation Accuracy	84.3%	68.5%
Character Accuracy	93.0%	88.5%

The accuracy metric is calculated as:

$$\text{Accuracy} = \frac{TN - EN}{TN} \times 100\% \quad (1)$$

where TN = total samples, EN = erroneous corrections.

4.2 Kimi LLM limitations

Experimental results on 100 test images processed by the Kimi model reveal significant performance limitations:

1. **Failure Rate:** 25% of images (25/100) were unparseable due to severe quality degradation (e.g., extreme blur or distortion).
2. **Error Rate:** 35% of images (35/100) yielded incorrect text extraction, including character misrecognition and punctuation omissions.
3. **Accuracy:** Only 40% of images (40/100) were successfully recognized with full fidelity to ground-truth text.

This suboptimal performance underscores the challenges of direct image-to-text processing with standalone LLMs. The hybrid framework—combining dedicated OCR engines (e.g., PP-OCRv4) for initial recognition and LLMs for post-correction—proves essential to mitigate such limitations.

4.3 Error typology case studies

Case 1 (Punctuation Error)

Input: 妈妈打了个哈欠：好久没走那么多路了！

Output: 妈妈打了个哈欠：“好久没走那么多路了！”

Analysis:

Error Type: Missing punctuation (missing quotation marks for direct speech)

Correction Mechanism:

The model recognized the absence of quotation marks in the input sentence, where the latter part is a direct speech. By utilizing contextual understanding, the model correctly inserted Chinese quotation marks around the quoted speech, restoring proper punctuation.

Case 2 (Character Substitution)

Input:会变成一个冷漠的人，或者情夫。

Output: 会变成一个冷漠的人，或者懦夫。

Analysis:

Error Type: Character shape error (incorrect use of“情夫”)

Correction Mechanism:

The model identified the inconsistency in the word choice (“情夫” vs. “懦夫”). Given the psychological context of the sentence, the model recognized that "情夫" (lover) was incorrect and substituted it with "懦夫" (coward), which better fits the context. The error was corrected based on semantic understanding and pretrained knowledge of common word usage.

4.3.1 Failure mode statistics

The residual error distribution after LLM correction is shown in **Table 2**.

Table 2. Residual error distribution after LLM correction:

Error Type	Frequency	Primary Cause
Mixed Punctuation	40.2%	EN/CN symbol similarity (e.g., , vs. ,)
Rare Glyph Errors	27.5%	Uncommon Chinese characters (e.g., 轅)
Character Omission Errors	32.3%	Poor image quality and insufficient contextual information

5. CONCLUSION

This study evaluates post-OCR correction frameworks on proprietary medical datasets, comparing the baseline performance of PP-OCrv4 with LLM-enhanced pipelines. Key findings include: Direct image-to-text processing using the Kimi model exhibited **25%** failure rates due to image quality issues, underperforming PP-OCrv4-based workflows. The GLM-4-Plus model achieved **17%** correction accuracy on low-confidence OCR outputs, delivering a **4.7%** system-wide accuracy improvement—significantly outperforming GLM-4-Flash (1.56% gain). We propose a standardized OCR correction framework: Initial Recognition deploying PP-OCrv4 for robust text extraction, LLM Refinement routing low-confidence outputs to GLM-4-Plus for contextual error correction, and Domain Adaptation fine-tuning models on task-specific data (e.g., medical terminology). This hybrid approach enables scalable, high-precision text recognition for critical applications like clinical documentation and legal archives.

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